Information Sheet

When Does Graph Structure Help in Multi-Agent Reinforcement Learning? The Right Edges, Not More Edges — Ali Jabbary and Kasra Ghanavati.

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1. What is the main claim of the paper? Why is this an important contribution to the autonomous agents and multi-agent systems literature?

The main claim is a **design rule with measured boundaries**: adding a coordination graph to value-factorised cooperative MARL helps *only* when the graph's edges match the task's information dependencies — the right edges, not more edges — and this is a property of the *topology*, not of any one coordination mechanism. Concretely: (i) when an agent must act on private information held by one specific, unobserved partner, a task-matched graph decisively outperforms all-to-all communication, a density-matched wrong graph, and a parameter-matched no-graph control (Cohen's $d \approx 5$, complete seed-level separation); (ii) the same result is reproduced by a deep coordination graph in the classical Guestrin / Kok–Vlassis lineage (DCG with max-plus action selection), which likewise collapses to floor on wrong or dense graphs — so the rule generalises across mechanism families; (iii) when coordination is convention-reducible or fully observed, the graph confers no benefit at matched capacity, and the commonly used unstabilised GCN operator is actively harmful — an optimisation pathology that stabilisation removes. The positive endpoint is scoped to the constructed archetypes on which it was measured: a pre-registered attempt to transfer it onto third-party MPE dynamics did not carry it there, and we report that attempt as the negative it was.

This matters to the AAMAS community because coordination graphs originate here, yet modern graph-MARL papers rarely separate the contribution of *structure* from the capacity and communication machinery that carries it. The paper supplies the missing controlled attribution — two isolators (a byte-identical no-graph control; a density-matched wrong-graph control) — and reconciles the field's conflicting positive and negative reports via the task's information structure.

2. What is the evidence you provide to support your claim? Be precise.

Controlled experiments on two constructed archetypes (pair-routing: distributed task allocation under partial observability with private goals; TokenMatch: referential signalling) plus MPE, with fixed encoder, mixer, optimiser, and parameter count across arms:

- **Positive endpoint.** Task-matched graph vs. all-to-all / wrong-graph / no-graph at matched parameters and observations: $d \approx 5$ at 30k steps, growing to $d \approx 12$ at 150k (12 seeds, Welch + Holm and Mann–Whitney, $p_{\text{Holm}} < 10^{-4}$, complete seed-level separation). Replicated out-of-harness on TokenMatch (∼96% of maximum vs. rivals far below). Learned attention (GAT / DGN) recovers the signal only on the right graph, at both tested budgets.
- **Classical baseline** (pre-registered, timestamp-verifiable). A steelman deep coordination graph (DCG, max-plus) *matches* the 1-hop positive on the true graph (n.s. vs. GNN-QMIX) and collapses to floor on wrong / dense graphs ($d > 12$, $p_{\text{Holm}} < 10^{-10}$) — establishing mechanism-agnostic structure-dependence. On a 2-hop relay task, neither mechanism routes above a no-communication baseline at 150k steps; reported in full as a pre-registered negative. A pre-registered addendum completes that relay cell with its two missing within-GNN controls: the task-matched, wrong and all-to-all graph arms are mutually indistinguishable there and none clears the no-communication control — a within-GNN null.
- **Attempted transfer to third-party dynamics** (pre-registered, timestamp-verifiable), reported as a negative. We composed an MPE `simple_reference` vehicle whose private-partner-goal structure is native, disabled its own communication channel, and froze the arms, metric, contrast family, oracle gate and decision branches before running. The oracle gate passed (+30.06 over the no-graph control, $d = 2.16$, 12 seeds, 150k steps), so the task is solvable when the information is routed — but all three graph arms diverged during training and finished below both the no-graph control (task-matched graph −35.96, $d = -1.73$, $p_{\text{Holm}} = 1.0 \times 10^{-3}$, MWU $= 7.3 \times 10^{-4}$) and the random-policy floor. We publish this as the pre-registered branch it landed on, characterise the instability without claiming a mechanism for it, and rescope the positive endpoint to the constructed archetypes on which it was measured.
- **Negative endpoint.** At full information the graph confers no benefit over the byte-identical no-graph control; the unstabilised operator's penalty grows with team size and recurs on MPE `simple_spread` (directionally on LBF); residual + LayerNorm stabilisation restores parity on both tested topologies.
- **Protocol.** $n = 10$–12 seeds per confirmatory cell, Holm-corrected Welch t and Mann–Whitney across the pre-declared contrast family, effect sizes and CIs throughout; three pre-registrations are timestamp-verifiable in the public repository's git history (committed, with their decision rules and budget freezes, strictly before the data they govern, in both wall-clock time and git ancestry); all code, configurations, per-run manifests, and analysis scripts are released.

3. What papers by other authors make the most closely related contributions, and how is your paper related to them?

- **Kok & Vlassis**, *JMLR* 7:1789–1828 (2006), and Sparse Cooperative Q-learning, *ICML* 2004: established sparse-beats-dense for tabular payoff propagation over a known coordination graph. We test this lineage directly via its neural descendant and show the sparse-structure rule survives deep function approximation under partial observability and private state — while adding the capacity-matched and wrong-graph controls the classical results did not run.
- **Guestrin, Koller & Parr** (*NIPS* 2001; *ICML* 2002); **Guestrin et al.**, *JAIR* 19:399–468 (2003): origin of the coordination-graph formalism. Our environment's edge-factored reward is a pairwise-factored payoff in exactly this sense; our contribution is empirical attribution of the structure's value inside deep value-factorisation.
- **Böhmer, Kurin & Whiteson**, *ICML* 2020 (Deep Coordination Graphs): the modern deep CG; we run it as our pre-registered classical baseline.
- **Rashid et al.** (QMIX, *ICML* 2018 / *JMLR* 2020) and the value-decomposition line (VDN, *AAMAS* 2018): the factorisation backbone our harness holds fixed.
- **Agarwal et al.**, *NeurIPS* 2021 (statistical precipice): the evaluation-rigor protocol our statistics follow.
- **Panait & Luke**, *JAAMAS* 11(3):387–434 (2005): the cooperative-MAL survey whose framing of coordination our boundary results refine.

4. Have you published parts of your paper before, for instance in a conference? If so, give details.

No part of this paper has been published in any archival venue, conference, or workshop, and no part is under review elsewhere. An earlier version was submitted to and rejected by TMLR (desk) and JAIR (desk, without review); neither constitutes a publication, and the manuscript has been substantially revised and extended since (the capacity-matched and wrong-graph controls, the 150k robustness study, and the pre-registered classical coordination-graph baseline are all new). There is no arXiv preprint at submission time.